

1. What is the main purpose of a DNS server?

- A) Assign MAC addresses
- B) Assign IP addresses
- C) Resolve domain names to IP addresses ☒
- D) Monitor traffic on the network

☒ **Answer:** C

Explanation: DNS translates human-friendly domain names to IP addresses.

2. Which Windows Server role provides automatic IP address configuration to clients?

- A) DNS
- B) IPAM
- C) DHCP ☒
- D) Active Directory

☒ **Answer:** C

Explanation: DHCP automates IP address assignment.

3. Which record type in DNS maps a domain name to an IPv4 address?

- A) PTR
- B) CNAME
- C) A
- D) AAAA

Answer: C

Explanation: An "A" record maps domain names to IPv4 addresses.

4. What does IPAM stand for?

- A) Internet Protocol Address Manager
- B) Internal Protocol Address Mapping
- C) IP Address Management
- D) IP Access Management

Answer: C

Explanation: IPAM is a centralized system for managing IP addresses.

5. What is the default DHCP lease duration on a Windows Server?

- A) 8 hours
- B) 1 hour
- C) 1 day ☒
- D) 7 days

☒ **Answer:** C

Explanation: By default, leases are granted for 8 days; one day is a typical choice during config.

6. Which tool is used to install server roles in Windows Server?

- A) Task Scheduler
- B) Server Manager ☒
- C) IPAM Console
- D) Registry Editor

☒ **Answer: B**

Explanation: Server Manager is the GUI tool for installing roles and features.

7. A DNS PTR record is used for:

- A) Forward lookup
- B) Reverse lookup ☒
- C) DHCP lease
- D) IPAM logging

☒ **Answer: B**

Explanation: PTR records resolve IP addresses to hostnames (reverse DNS).

8. What is a DHCP reservation?

- A) A blocked IP
- B) An IP assigned randomly
- C) An IP permanently leased to a device ☒
- D) A DNS record

☒ **Answer:** C

Explanation: A reservation ties a specific IP to a MAC address.

9. Which type of DNS zone allows for read-write access?

- A) Secondary
- B) Stub
- C) Primary ☒
- D) Reverse

☒ **Answer:** C

Explanation: Only primary zones can be updated directly.

10. Where is the IPAM feature installed from?

- A) Windows Update
- B) Command Prompt

- C) Add Roles and Features Wizard ☒
- D) Registry Editor

☒ **Answer:** C

Explanation: IPAM is installed like other roles using Server Manager.

☒ Medium (20 Questions)

11. What port does DNS use by default?

- A) TCP 25
- B) TCP/UDP 53 ☒
- C) UDP 67
- D) TCP 80

☒ **Answer:** B

Explanation: DNS uses port 53 for both TCP (zone transfers) and UDP (queries).

12. What command can you use to flush a DNS cache on a client?

- A) ipconfig /registerdns
- B) netsh int ip reset
- C) ipconfig /flushdns ☒
- D) nslookup /clearcache

☒ **Answer: C**

Explanation: This clears the resolver cache on a Windows client.

13. Which of the following is NOT a valid DHCP scope option?

- A) Default Gateway
- B) Subnet Mask
- C) Hostname ☒
- D) DNS Server

☒ **Answer: C**

Explanation: Hostname is not configured in DHCP scope options.

14. DHCP and DNS can be integrated to provide:

- A) ARP spoofing
- B) Network printing
- C) Dynamic DNS updates ☒
- D) VPN tunneling

☒ **Answer:** C

Explanation: DHCP can dynamically update DNS records for clients.

15. A stub zone contains:

- A) All resource records
- B) Only records necessary for forwarding queries ☒
- C) Just PTR records
- D) Secondary zone data

☒ **Answer:** B

Explanation: Stub zones include SOA, NS, and A records of authoritative DNS servers.

16. Which Windows PowerShell command installs the DHCP role?

- A) `Install-WindowsFeature DHCP -IncludeManagementTools` ☒
- B) `Enable-DHCPServer`
- C) `Add-DHCPRole`
- D) `New-DHCPServer`

☒ **Answer:** A

Explanation: That is the correct PowerShell syntax.

17. What IPAM discovery method uses Group Policy to manage DHCP/DNS servers?

- A) Automatic
- B) Manual
- C) GPO-based provisioning ☒
- D) Static assignment

☒ **Answer:** C

Explanation: GPO provisioning allows IPAM to manage servers using domain policies.

18. What is the primary function of a DHCP relay agent?

- A) Encrypt IP traffic
- B) Translate IPv6 addresses
- C) Forward DHCP requests across subnets ☒
- D) Assign hostnames

☒ **Answer:** C

Explanation: Relay agents enable DHCP clients on different subnets to reach a DHCP server.

19. Which of these tasks can you perform using the IPAM console?

- A) Install device drivers
- B) Add DNS records

- C) Track IP usage across subnets ☒
- D) Patch Windows updates

☒ **Answer:** C

Explanation: IPAM tracks and audits IP address usage.

20. What does a DHCP lease contain?

- A) MAC address, IP, and TTL ☒
- B) Hostname and TTL only
- C) Subnet and DNS only
- D) Scope ID and VLAN

☒ **Answer:** A

Explanation: A lease includes client MAC, assigned IP, and lease duration (TTL).

21. In DHCP, what happens when the lease duration expires?

- A) The IP is assigned to another client immediately
- B) The client releases the IP
- C) The IP address becomes invalid until reassigned
- D) The client must renew the lease ☒

☒ **Answer: D**

Explanation: DHCP clients must renew expired leases to retain their IP.

22. Which of the following tools is best suited for real-time DHCP monitoring?

- A) Event Viewer
- B) Performance Monitor
- C) DHCP MMC Console ☒
- D) IPAM PowerShell

☒ **Answer: C**

Explanation: The DHCP Management Console offers real-time lease and activity views.

23. What happens if you do not authorize a DHCP server in Active Directory?

- A) It continues functioning normally
- B) It shuts down
- C) It does not assign IPs in a domain ☒
- D) It blocks DNS updates

☒ **Answer:** C

Explanation: Unauthorized DHCP servers do not provide IPs in AD-integrated networks.

24. IPAM primarily helps prevent:

- A) DNS cache poisoning
- B) IP conflicts ☒
- C) MAC spoofing
- D) Brute-force attacks

☒ **Answer:** B

Explanation: IPAM prevents duplicate IP assignments and allows central tracking.

25. What type of zone allows DNS replication but disallows updates?

- A) Primary
- B) Secondary ☒
- C) Stub
- D) Forward lookup

☒ **Answer:** B

Explanation: Secondary zones are read-only replicas.

26. What PowerShell command displays all DHCP leases?

- A) Get-DHCPLease
- B) Get-DhcpServerv4Lease ☒
- C) Show-DhcpLeases
- D) List-DHCP

☒ **Answer:** B

Explanation: This command lists all active leases.

27. Which log location stores DHCP server events?

- A) %systemroot%\System32\DHCP\DhcpSrvLog-*.log
- B) C:\Windows\Logs\Security
- C) %temp%\dhcp.log
- D) C:\Program Files\DHCP\logs

Answer: A

Explanation: These are the standard DHCP logs.

28. Which record should be configured to allow reverse lookups?

- A) PTR
- B) MX
- C) A
- D) CNAME

Answer: A

Explanation: PTR records are used for reverse name resolution.

29. Which of the following does IPAM integrate with?

- A) IIS
- B) WINS
- C) DNS and DHCP ☒
- D) SMB

☒ **Answer: C**

Explanation: IPAM centrally manages DNS and DHCP servers.

30. In a large enterprise, what's a major advantage of IPAM?

- A) Reduced DNS queries
- B) Lower bandwidth use
- C) Centralized IP tracking and auditing ☒
- D) Faster OS boot times

☒ **Answer: C**

Explanation: IPAM helps avoid conflicts and ensures accountability in IP management.

☒ Hard (10 Questions)

31. Which IPAM database table stores relationships between address blocks and DNS zones?

- A) IPAMZoneLink
- B) IpamIPAddressRecord
- C) IpamRangeToZoneLink ☒
- D) IPAMZoneIndex

☒ **Answer:** C

Explanation: This table links IP address ranges with zones.

32. How does DNS aging/scavenging work?

- A) Removes stale A records after fixed days ☒
- B) Resets PTR records automatically
- C) Converts static records to dynamic
- D) Deletes all zones periodically

☒ **Answer: A**

Explanation: Aging/scavenging removes outdated DNS entries.

33. What GPO settings does IPAM configure during provisioning?

- A) DHCP lease time
- B) DNS forwarders
- C) Firewall rules, scheduled tasks, and event forwarding ☒
- D) Hostname changes

☒ **Answer: C**

Explanation: These enable remote management.

34. What is the minimum IP range needed for a scope supporting 120 hosts?

- A) /27
- B) /25 ☒
- C) /26
- D) /24

☒ **Answer:** B

Explanation: /25 provides 128 addresses, supporting 120 usable hosts.

35. Which protocol does DHCP use?

- A) TCP
- B) UDP ☒
- C) ICMP
- D) SNMP

☒ **Answer:** B

Explanation: DHCP operates over UDP ports 67 (server) and 68 (client).

36. You need to configure DNS for conditional forwarding. What is required?

- A) Reverse lookup zone
- B) PTR records
- C) FQDN of target domain ☒
- D) MAC address of server

☒ **Answer: C**

Explanation: Conditional forwarding uses FQDNs.

37. What is the function of the DNS "SOA" record?

- A) Resolves aliases
- B) Tracks DHCP leases
- C) Defines zone authority and settings ☒
- D) Manages reverse lookup

☒ **Answer: C**

Explanation: SOA includes serial number, TTL, and primary NS info.

38. What happens if a DHCP scope's lease pool is exhausted?

- A) Clients use APIPA addresses ☒
- B) Clients retry the request
- C) Clients shut down
- D) Server assigns duplicate IPs

☒ **Answer:** A

Explanation: Clients assign themselves 169.254.x.x APIPA addresses.

39. How does IPAM ensure integrity of IP assignments?

- A) Encrypts leases
- B) Blocks rogue servers
- C) Correlates DNS and DHCP data ☒
- D) Removes PTR records

☒ **Answer:** C

Explanation: IPAM cross-references DNS/DHCP data.

40. What happens during the DHCP "Rebinding" state?

- A) Client discovers a DHCP server
- B) Server renews the lease
- C) Client tries other DHCP servers
- D) Server deletes the lease

☒ **Answer:** C

Explanation: In rebinding, clients contact any DHCP server after failed renewal.